



Product Requirements Document (PRD) — PrintEve v1.0

PrintEve — Full Platform v1.0

| **Version:** 1.0 · **Status:** Draft · **Stage:** Greenfield · **Last Updated:** June 2026

1. Executive Summary

PrintEve is an on-demand custom printing platform that connects customers who want personalised print products (Stickers, posters, Banners, Pamphlets, batches etc) with a network of verified printer vendors who fulfil those orders.

The platform operates across four surfaces — a customer website, a customer mobile app, an internal admin dashboard, and a dedicated printer vendor portal — all powered by a single Bun API backend at `api.printeve.in`.

V1 delivers a complete end-to-end platform: customers browse, customise, and order; printers receive, fulfil, and dispatch; admins manage the full operation from one place. PrintEve takes a platform margin on each fulfilled order.

2. Problem Statement

For customers: Getting custom printed products today means finding a local printer, negotiating pricing, sending files over WhatsApp, and hoping for quality. There is no reliable, branded, digital-first experience for ordering custom print goods online.

For printer vendors: Small and mid-size print shops have idle capacity but no scalable channel for consistent order volume. They rely on walk-ins and word of mouth with no digital job management tooling.

For the business: A clear gap exists for a platform that aggregates print demand and routes it efficiently to a vendor network — taking a margin on each

order while giving both sides a superior experience.

3. Goals & Success Metrics

Business goals

- Launch with at least 3 onboarded printer vendors before public launch
- Process first 100 orders within 30 days of launch

Product goals

- Customer: landing page → placed order in under 5 minutes
 - Printer: accept and update a job status in under 2 minutes
 - Admin: assign an order to a printer within 1 minute of placement
-

4. Users & Personas

Customer (Web + App)

Individuals and small businesses ordering custom printed products.

- **Needs:** easy product discovery, transparent pricing, real-time order tracking, reliable quality
- **Pain points today:** no trusted online destination, unclear pricing, no tracking after ordering, inconsistent quality

Admin ([admin.printeve.in](#))

PrintEve internal operations team.

- **Needs:** full visibility across all orders, users, and vendors; ability to manually assign print jobs to the right vendor; product catalog management; revenue reporting; platform settings control

Printer Vendor ([printer.printeve.in](#))

External print shop owners or operators fulfilling orders.

- **Needs:** clear incoming job view with all files and specs, simple accept/reject flow, status update tools, proof upload, earnings visibility
- **Pain points today:** all job management over WhatsApp, no proof-of-completion workflow, unpredictable order volume

5. Scope — V1

In Scope

- Customer web app — `printeve.in`
- Customer mobile app — iOS + Android (login mandatory)
- Admin dashboard — `admin.printeve.in`
- Printer vendor portal — `printer.printeve.in`
- Single Bun API — `api.printeve.in` with 4 namespaces: `/v1/web`, `/v1/app`, `/admin`, `/printer`
- Product catalog with categories and variants
- Custom design file upload (PNG, PDF, AI — max 3GB)
- Cart and checkout with Razorpay payment
- Order management and live status tracking
- Manual print job assignment (admin assigns to vendor)
- Print job fulfillment workflow
- Email notifications via Resend
- Push notifications via FCM (mobile app)
- Admin reporting and analytics
- Printer vendor onboarding and profile management
- Customer data anonymised for printer vendors (order ID + area only)

Out of Scope — Post-V1

- In-browser design editor
- Multi-vendor cart
- Auto-assignment of jobs by location or capacity
- Guest checkout on mobile app
- Customer reviews and ratings
- Loyalty points / rewards

- International shipping
- White-label printer portal
- Public API for third-party integrations

6. Features & Requirements

Priority: **P0** = must ship · **P1** = should ship · **P2** = nice to have

6.1 Customer Web (printeve.in)

Authentication

Requirement	Priority
Register with name, email, and password	P0
Login → HttpOnly session cookie (7-day sliding expiry)	P0
Logout clears session server-side	P0
Password reset via email OTP	P1
Google OAuth login	P2

Product Catalog

Requirement	Priority
Homepage with featured products and categories	P0
Browse by category with filters (price, category)	P0
Product listing — paginated, filterable	P0
Product detail — images, description, variants, pricing	P0
Upload custom design file (PNG, PDF, AI — max 3GB)	P0
Product search by name / keyword	P1
Recently viewed products	P2

Cart

Requirement	Priority
Add product with variant and quantity to cart	P0

Requirement	Priority
Update quantity or remove items	P0
Cart persists across sessions (linked to account)	P0
Cart shows subtotal, tax, and total	P0
Apply coupon code	P1

Checkout

Requirement	Priority
Select or add delivery address	P0
Order summary shown before payment	P0
Pay via Razorpay — UPI, card, netbanking, wallets	P0
On success → order created, confirmation email sent	P0
On failure → error shown with retry option	P0
Razorpay webhook handles async confirmation (idempotent)	P0

Orders

Requirement	Priority
View order history	P0
View full order detail — items, status, address, payment	P0
Live order status (refreshes on page load)	P0
Email notification on each status change	P1
Download invoice as PDF	P2

Account

Requirement	Priority
View and update profile	P0
Address book — add, edit, delete, set default	P0
Change password	P1

6.2 Customer Mobile App (iOS + Android)

Login is mandatory. Guest checkout is not supported in V1.

Authentication

Requirement	Priority
Register with name, email, password	P0
Login → access token (15 min) + refresh token (30 days)	P0
Silent token refresh on expiry	P0
Logout invalidates refresh token server-side	P0
Biometric login after first login (FaceID / fingerprint)	P2

Product Catalog

Requirement	Priority
Home screen — featured products and categories	P0
Browse by category with filters	P0
Product detail — images, variants, pricing	P0
Upload custom design from camera roll or files	P0
Product search by keyword	P1

Cart & Checkout

Requirement	Priority
Add, update, remove cart items	P0
Cart persists across app sessions	P0
Select or add delivery address at checkout	P0
Pay via Razorpay — UPI, card, wallets	P0
Order confirmation screen on success	P0
Coupon code input	P1

Orders & Tracking

Requirement	Priority
Order history screen	P0
Order detail with live status	P0

Requirement	Priority
Push notification on every order status change	P0
FCM device token registered on login	P0

Profile

Requirement	Priority
View and edit profile	P0
Address book management	P0
Notification preferences	P1

6.3 Admin Dashboard (admin.printeve.in)

Authentication & Access

Requirement	Priority
Admin login → JWT (separate secret from all other namespaces)	P0
Portal IP-restricted at Nginx level	P0
Every admin action audit-logged — who, what, when, record ID	P0

Dashboard

Requirement	Priority
KPIs: revenue today, orders today, new users, pending print jobs	P0
Revenue chart — daily / weekly / monthly toggle	P0
Orders by status breakdown chart	P1
Top selling products widget	P1

Order Management

Requirement	Priority
View all orders — filter by status, date range, customer	P0
View full order detail	P0
Manually update order status	P0

Requirement	Priority
Manually assign order to a printer vendor → creates PrintJob	P0
Issue full or partial refund	P1
Export orders to CSV	P2

Product Management

Requirement	Priority
View all products with active/inactive status	P0
Create product — name, description, category, variants, pricing, images	P0
Edit existing product	P0
Delete / deactivate product	P0
Manage categories — create, edit, delete, reorder	P0

User, Printer & Coupon Management

Requirement	Priority
View all customers — searchable and filterable	P0
View customer profile and full order history	P0
Block / unblock a customer account	P1
View all printer vendor accounts	P0
Create printer account — generates login credentials	P0
Edit printer account details	P0
Deactivate / remove printer vendor	P1
Create coupon — code, flat/%, min order, expiry, usage limit	P0
View and delete coupons	P0

Reports & Settings

Requirement	Priority
Revenue report — by day / week / month / custom range	P0
Order report — volume, status breakdown, AOV	P1
Product performance — top sellers, revenue by product	P1

Requirement	Priority
Configure tax rate (%)	P0
Configure flat or tiered shipping charges	P0
Free shipping threshold	P1

6.4 Printer Vendor Portal (printer.printeve.in)

Customer data is anonymised for all printer vendors. Jobs show order ID and delivery area only — no customer name or full address is ever exposed.

Authentication

Requirement	Priority
Printer login → JWT (separate secret)	P0
Portal IP-restricted at Nginx level	P0
Access token: 12h · Refresh token: 14 days	P0

Dashboard

Requirement	Priority
Stats: pending jobs, in-progress, completed, earnings this month	P0
Recent job activity feed	P1

Job Management

Requirement	Priority
View all jobs assigned to this printer only (hard-scoped in API)	P0
Job detail — product specs, quantity, design files, delivery area (anonymised)	P0
Accept a job	P0
Reject a job with a mandatory reason	P0
Update status: Printing → Ready → Dispatched	P0
Upload proof photo on job completion	P0
View completed and rejected job history	P1

Profile

Requirement	Priority
View and update business name, address, contact details	P0
View account status — active / suspended	P1

7. Order Lifecycle

Order Status Flow

Status	Description
PENDING	Order placed, payment initiated via Razorpay
CONFIRMED	Payment confirmed via webhook — customer notified
PRINTING	Admin assigns to printer, printer accepts — job in progress
READY	Printer marks job ready, uploads proof photo
SHIPPED	Printer marks dispatched — customer notified
DELIVERED	Admin marks delivered — customer notified
CANCELLED	Cancelled before printing begins
REFUNDED	Refund issued by admin

Print Job Status Flow

Status	Description
ASSIGNED	Job created and manually assigned to a printer by admin
ACCEPTED	Printer accepts the job
IN_PROGRESS	Printer is actively working on the job
READY	Job complete, awaiting dispatch
DISPATCHED	Job shipped to customer
REJECTED	Printer rejects — admin must re-assign manually

Note: Auto-assignment by location or capacity is planned for a post-V1 release.

8. Notification Strategy

Event	Customer Email	Customer Push	Admin	Printer
Order placed	✓	✓	—	—
Payment failed	✓	✓	—	—
Order confirmed	✓	✓	—	—
Job assigned to printer	—	—	—	✓ Email
Printer accepts job	—	—	✓	—
Printer rejects job	—	—	✓	—
Order status: Printing	✓	✓	—	—
Order status: Shipped	✓	✓	—	—
Order status: Delivered	✓	✓	—	—
Refund issued	✓	✓	—	—

9. Non-Functional Requirements

Performance

- API p95 response time < 300ms under normal load
- Product listing page loads in < 2 seconds
- Design file uploads support up to 2GB

Security

- All traffic over HTTPS (TLS 1.2+)
- JWT secrets isolated per namespace — a customer token is useless against admin routes
- Admin and printer portals IP-restricted at Nginx level
- Passwords hashed with bcrypt (min 12 rounds)
- CSRF protection on all web session mutating requests
- Zod validation on every endpoint before controller logic runs

- Rate limiting on all public endpoints

Reliability

- Razorpay webhooks must be idempotent — duplicate events must not create duplicate orders
- Print job assignment must be atomic — a job cannot be assigned to two printers simultaneously
- Refresh token rotation — old token invalidated on each use

Scalability

- Stateless API (except web sessions via Redis) — horizontally scalable
- File storage on Cloudflare R2 — never local disk
- DB connection pooling via DrizzleORM

Observability

- Request logging on all endpoints
- Admin audit log for every admin action
- Printer action log for every job update
- Error tracking via Sentry or equivalent

10. Technical Constraints

Constraint	Detail
Runtime	Bun (not Node.js)
Language	TypeScript throughout
ORM / DB	DrizzleORM + PostgreSQL (prod) / SQLite (dev)
Session Store	Redis for web sessions
Validation	Zod — runs before every controller method
Auth	JWT via jose — three separate secrets (app / admin / printer)
File Storage	Cloudflare R2 — no direct SDK calls in controllers
Email	Resend
Push	Firebase Cloud Messaging (FCM)
Payment	Razorpay

Constraint	Detail
Proxy	Nginx — reverse proxy + IP whitelist for <i>/admin/</i> and <i>/printer/</i>

Hard rules:

- No business logic in controllers — services only
- No direct DB calls outside the models/services layer
- No shared JWT secrets across namespaces
- All file operations through UploadService
- Zod validation must run before every controller method
- Printer API never returns customer name or full address — order ID + area only

11. Milestones

Phase	Scope
1	Core framework — Bun server, Router, Middleware chain, DB
2	Models — User, Product, Order, Cart, PrintJob, etc.
3	Shared services — Auth, Order, Cart, Payment, Print, Upload, Notification
4	<i>/v1/web</i> namespace — full customer web API
5	<i>/v1/app</i> namespace — full mobile app API
6	<i>/admin</i> namespace — full admin dashboard API
7	<i>/printer</i> namespace — full printer portal API
8	Frontend — printeve.in (customer web)
9	Frontend — admin.printeve.in
10	Frontend — printer.printeve.in
11	Mobile app — iOS + Android
12	QA, load testing, security audit
13	Nginx, SSL, Redis, production deployment
14	Soft launch — 3 onboarded printer vendors